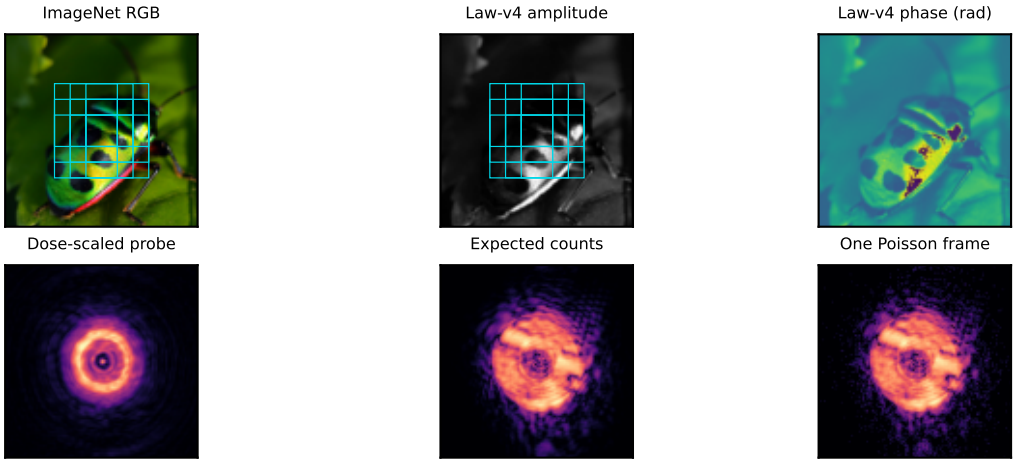
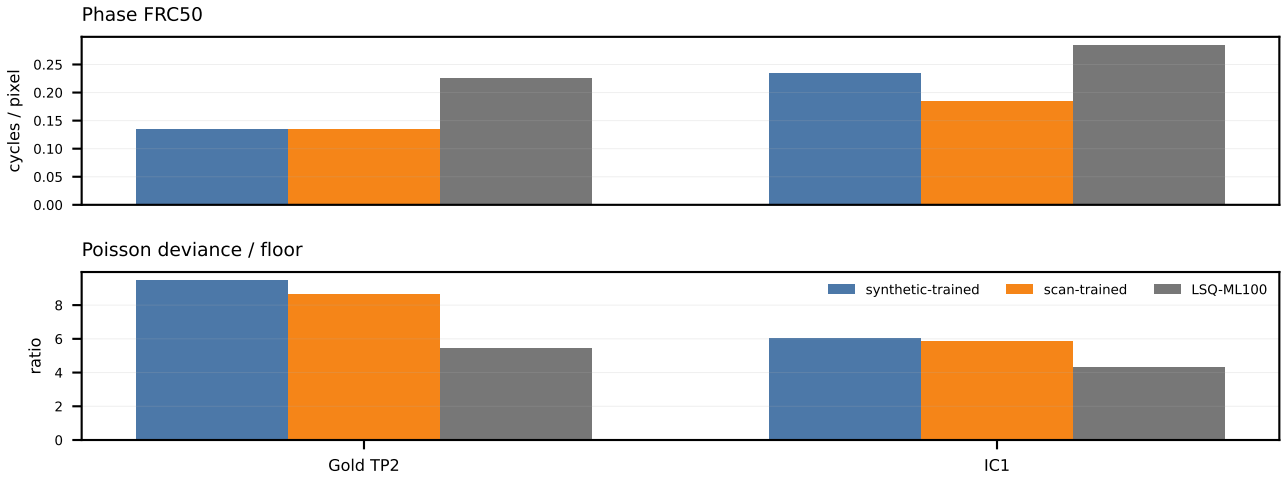
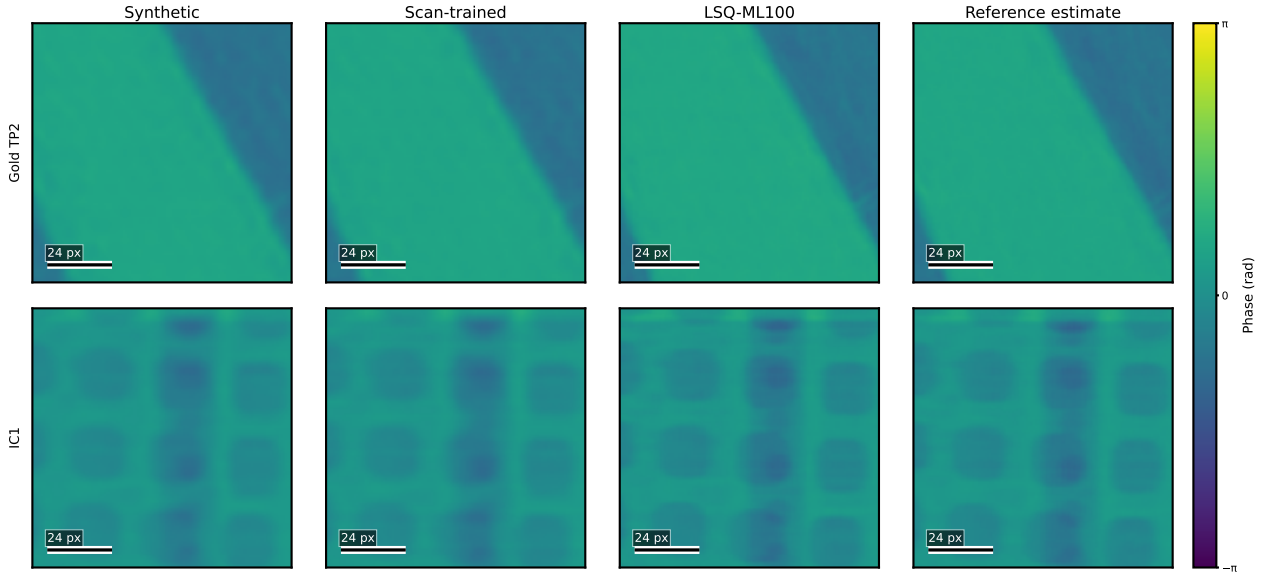
Supplementary: Synthetic generation and transfer to measured scans

haematite; 8 keV; 8.15 µm; excess absorption 2.03 · 1,852,972 photons/frame; 9 linked patches; 9,000 synthetic training frames







Synthetic checkpoint selected on synthetic validation; scan-trained checkpoint selected on high-half spatial validation (not an independent test). Shared anchor, common illuminated mask and border; LSQ-ML is rescored on native objects.